

Web Technologies Assignment-1

1.Scenario: A newly established educational institution plans to launch its official website. Before

developing the website, the management wants to understand the fundamental concepts of the World Wide Web.

Task: Create a simple HTML webpage explaining the following concepts with suitable illustrations

or diagrams:

- Internet Overview
- Fundamental Computer Network Concepts
- URL and its Structure
- Domain Name
- Web Browsers and Web Servers
- Working Principle of a Website
- Difference between Client-side and Server-side Scripting

Program:

index.html:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Fundamentals of the World Wide Web</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <h1>Fundamentals of the World Wide Web</h1>
    <p>Understanding the Internet and Web Technologies</p>
  </header>

  <nav>
```

```
<a href="#internet">Internet</a>
<a href="#url">URL</a>
<a href="#browser">Browser & Server</a>
<a href="#scripting">Scripting</a>
</nav>

<main>

<section id="internet">
  <h2>Internet Overview</h2>
  <p>
    The Internet is a global network of interconnected
    computers and devices used for communication and
    sharing information.
  </p>
</section>

<section id="network">
  <h2>Network Concepts</h2>
  <p>
    A computer network connects devices using protocols
    such as TCP/IP to exchange data.
  </p>
</section>

<section id="url">
  <h2>URL Structure</h2>
  <p>
    Example:
    https://www.example.com/index.html
  </p>
</section>

<section id="browser">
  <h2>Web Browser and Web Server</h2>
  <p>
    A web browser sends requests to a web server,
    and the server responds with the requested webpage.
  </p>
</section>

<section id="working">
  <h2>How a Website Works</h2>
```

```
<ol>
  <li>User enters a URL.</li>
  <li>Browser sends a request to the server.</li>
  <li>Server processes the request.</li>
  <li>Webpage is returned to the browser.</li>
</ol>
</section>
```

```
<section id="scripting">
  <h2>Client-side vs Server-side Scripting</h2>
```

```
  <p>
    Client-side scripting runs in the browser,
    while server-side scripting runs on the web server.
  </p>
</section>
```

```
</main>
```

```
<footer>
  <p>Web Technology Assignment - 2026</p>
</footer>
```

```
</body>
</html>
```

Style.css:

```
body {
  font-family: Arial, sans-serif;
  margin: 0;
  background: #f4f7fb;
  color: #222;
}
```

```
header {
  background: #123b68;
  color: white;
  text-align: center;
  padding: 35px;
}
```

```
nav {
  background: #102d4a;
```

```
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
}

main {
    width: 85%;
    margin: 30px auto;
}

section {
    background: white;
    padding: 25px;
    margin-bottom: 20px;
    border-radius: 12px;
}

h2 {
    color: #1769aa;
}

footer {
    background: #123b68;
    color: white;
    text-align: center;
    padding: 15px;
}
```

Output:

[Explore Concepts ↗](#)

WEB FUNDAMENTALS

How does the Web actually work?

The World Wide Web connects users, browsers, servers and information through a collection of technologies and communication protocols.

01 INTERNET

Internet Overview

The Internet is a global network of interconnected computers and devices that communicate with each other using standard communication protocols such as TCP/IP.

```
graph LR; Computer[Computer  
User Device] -- DATA --> Internet[Internet  
Global Network]; Internet -- DATA --> Mobile[Mobile  
User Device];
```

02 NETWORKING

Fundamental Computer Network Concepts

A computer network is a collection of interconnected devices that communicate and share resources. Important concepts include nodes, routers, IP addresses, protocols and data transmission.

```
graph LR; Client[Client  
Requests data] --> Router[Router  
Forwards data]; Router --> Server[Server  
Provides data];
```

NODE A device connected to a network.	ROUTER Forwards data between networks.	IP ADDRESS Identifies a device on a network.	PROTOCOL Rules used for communication.
---	--	--	--

03 ADDRESSING

URL and its Structure

A URL (Uniform Resource Locator) is the address used to locate a resource on the Internet.

<https://www.example.com/admission/index.html>

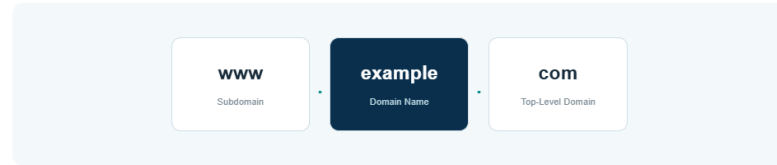
01 Protocol https://	02 Domain www.example.com	03 Path /admission/	04 Resource index.html
--------------------------------	-------------------------------------	-------------------------------	----------------------------------

04

IDENTIFICATION

Domain Name

A domain name is a human-readable address used to identify a website on the Internet. It is easier to remember than a numerical IP address.

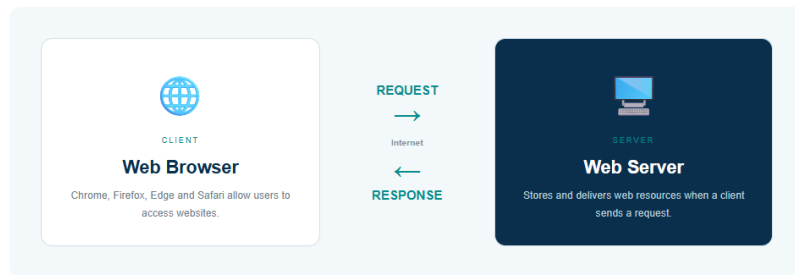


05

WEB INFRASTRUCTURE

Web Browsers and Web Servers

A web browser is software used to access and display web pages. A web server stores and delivers web resources to clients when requested.

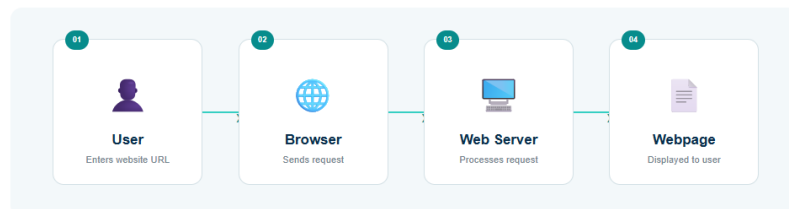


06

PROCESS

Working Principle of a Website

When a user enters a website address, the browser sends a request to the web server. The server processes the request and sends the required resources back to the browser. The browser then displays the webpage.





Explanation of Code

- The <header> section displays the main title and introduction of the webpage.
- The <nav> element creates navigation links to different sections of the webpage.
- The <section> elements organize the content into topics such as Internet, URL, Browser & Server, Website Working, and Scripting.
- The and elements display the steps involved in how a website works.
- The <link> tag connects the HTML page with the external CSS file.
- CSS is used to style the webpage by adding colors, spacing, backgrounds, navigation styles, and section layouts.
- Comments such as <!-- Internet Overview --> are included in the HTML code to clearly identify and explain different parts of the program.

Conclusion:

The webpage successfully presents the fundamental concepts of the World Wide Web. It explains Internet concepts, URLs, browsers, servers, and scripting. HTML is used to structure the content, while CSS is used to improve its appearance. Thus, the basic working and components of web technology were demonstrated successfully.

2.Scenario: A college wants to digitize its student admission process.

Task: Design a responsive Student Admission Form using HTML5 and CSS3. Your webpage should

demonstrate:

- HTML Form Elements
- Different Input Types
- Media Elements (Image/Video/Audio)
- CSS3 Selectors
- Box Model
- Responsive User Interface

Program:

Index.html:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Admission Form</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="container">

    <h1>Student Admission Form</h1>
    <p>Welcome to our College</p>

    <form>

      <label>Full Name</label>
      <input type="text" placeholder="Enter your name">
```

```

<label>Email</label>
<input type="email" placeholder="Enter your email">

<label>Date of Birth</label>
<input type="date">

<label>Gender</label>
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female

<label>Course</label>
<select>
  <option>Select Course</option>
  <option>B.E Computer Science</option>
  <option>B.E AI & ML</option>
  <option>B.E ECE</option>
</select>

<label>Phone Number</label>
<input type="tel" placeholder="Enter phone number">

<label>Address</label>
<textarea placeholder="Enter your address"></textarea>

<button type="submit">Submit Application</button>

</form>

<h2>Admission Welcome</h2>

<audio controls>
  <source src="admission-welcome.mp3" type="audio/mpeg">
</audio>

<video controls>
  <source src="college-admission.mp4" type="video/mp4">
</video>

</div>

</body>
</html>

```

Style.css:

```
body {
  font-family: Arial, sans-serif;
  background: #eef4f8;
  margin: 0;
  padding: 30px;
}

.container {
  max-width: 800px;
  margin: auto;
  background: white;
  padding: 30px;
  border-radius: 15px;
  box-shadow: 0 8px 25px rgba(0,0,0,0.1);
}

h1 {
  color: #123b68;
  text-align: center;
}

p {
  text-align: center;
}

img {
  width: 100%;
  max-height: 250px;
  object-fit: cover;
  border-radius: 10px;
}

form {
  margin-top: 25px;
}

label {
  display: block;
  margin-top: 15px;
  font-weight: bold;
}
```

```
input,
select,
textarea {
  width: 100%;
  padding: 12px;
  margin-top: 6px;
  border: 1px solid #bbb;
  border-radius: 6px;
  box-sizing: border-box;
}

textarea {
  height: 100px;
}

button {
  margin-top: 20px;
  padding: 12px 25px;
  background: #1769aa;
  color: white;
  border: none;
  border-radius: 6px;
}

audio,
video {
  width: 100%;
  margin-top: 15px;
}

@media (max-width: 600px) {
  .container {
    padding: 20px;
  }

  body {
    padding: 15px;
  }
}
```

Output:

Student Admission Portal

College Admission Registration
Start your journey towards a successful future

Student Admission Form

Please enter your details carefully

Personal Information

Full Name

Enter your full name

Email Address

example@gmail.com

Phone Number

Enter 10 digit phone number

Date of Birth

dd-mm-yyyy

Gender

☐ Male ☐ Female ☐ Other

Academic Information

Select Course

-- Select Course --

Higher Secondary Percentage

Enter percentage

Address

Enter your complete address

Additional Information

Upload Photograph

Choose File

No file chosen

Portfolio / Website URL

https://example.com

☐ I declare that the information provided above is correct.

Submit Application

Reset

College Information

Explore our campus and admission information

Our College Campus



Our college provides modern classrooms, laboratories, library facilities and a student-friendly campus environment.

College Campus & Admission Video



Watch the video to learn more about our college campus and admission facilities.

Admission Welcome Message Video

0:00 / 2:23

Listen to the admission welcome message before completing your application.

© 2026 Student Admission Portal
Web Technology Assignment

Explanation of Code

- HTML5 form elements such as <input>, <select>, <textarea> and <button> are used to collect student details.
- Different input types such as **text, email, date, radio and telephone** are demonstrated.
- , <audio> and <video> elements are used to include multimedia content.
- CSS selectors are used to style headings, form elements, buttons and media.
- The **box model** is demonstrated using padding, margin, border and width properties.
- The @media rule makes the admission form responsive on smaller screens.

Conclusion

The Student Admission Form was successfully designed using HTML5 and CSS3. Different form controls and input types were used to collect student information. Audio, video and image elements were incorporated into the webpage. CSS styling and responsive design make the form user-friendly across different screen sizes.

3.Scenario: An online fashion store wants an attractive home page to improve customer engagement.

Task: Design the homepage using HTML5 and CSS3 by incorporating:

- Backgrounds and Borders
- Text Effects
- CSS Animations
- Multiple Column Layout
- User Interface Design Principles

Program:

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>StyleHub - Fashion Store</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
```

```

<header>
  <h1>STYLEHUB</h1>
  <p>Discover Your Perfect Style</p>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">Men</a>
  <a href="#">Women</a>
  <a href="#">Footwear</a>
  <a href="#">Accessories</a>
</nav>

<section class="hero">
  <h2>New Fashion Collection</h2>
  <p>Explore the latest trends and styles.</p>
  <button>Shop Now</button>
</section>

<section class="products">

  <div class="card">
    
    <h3>Men's Fashion</h3>
    <p>Modern and stylish outfits.</p>
  </div>

  <div class="card">
    
    <h3>Women's Fashion</h3>
    <p>Elegant styles for every occasion.</p>
  </div>

  <div class="card">
    
    <h3>Footwear</h3>
    <p>Comfortable and fashionable shoes.</p>
  </div>

  <div class="card">
    
    <h3>Accessories</h3>
    <p>Complete your perfect look.</p>
  </div>

```

```
</div>
```

```
</section>
```

```
</body>
```

```
</html>
```

Style.css:

```
body {  
    margin: 0;  
    font-family: Arial, sans-serif;  
    background: #f7f7f7;  
}
```

```
header {  
    background: #171717;  
    color: white;  
    text-align: center;  
    padding: 35px;  
}
```

```
header h1 {  
    font-size: 38px;  
    letter-spacing: 5px;  
}
```

```
nav {  
    background: #222;  
    text-align: center;  
    padding: 15px;  
}
```

```
nav a {  
    color: white;  
    text-decoration: none;  
    margin: 20px;  
}
```

```
.hero {  
    text-align: center;  
    padding: 60px;  
    background: linear-gradient(135deg, #ffe0e8, #fff);  
}
```

```

.hero h2 {
  font-size: 40px;
  text-transform: uppercase;
}

button {
  padding: 12px 25px;
  border: none;
  border-radius: 25px;
  background: #e91e63;
  color: white;
}

.products {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 25px;
  padding: 40px;
}

.card {
  background: white;
  border: 1px solid #ddd;
  border-radius: 15px;
  overflow: hidden;
  text-align: center;
  animation: appear 1s ease;
}

.card img {
  width: 100%;
  height: 230px;
  object-fit: cover;
}

.card h3 {
  font-size: 22px;
}

.card:hover {
  transform: translateY(-8px);
  transition: 0.3s;
}

```

```

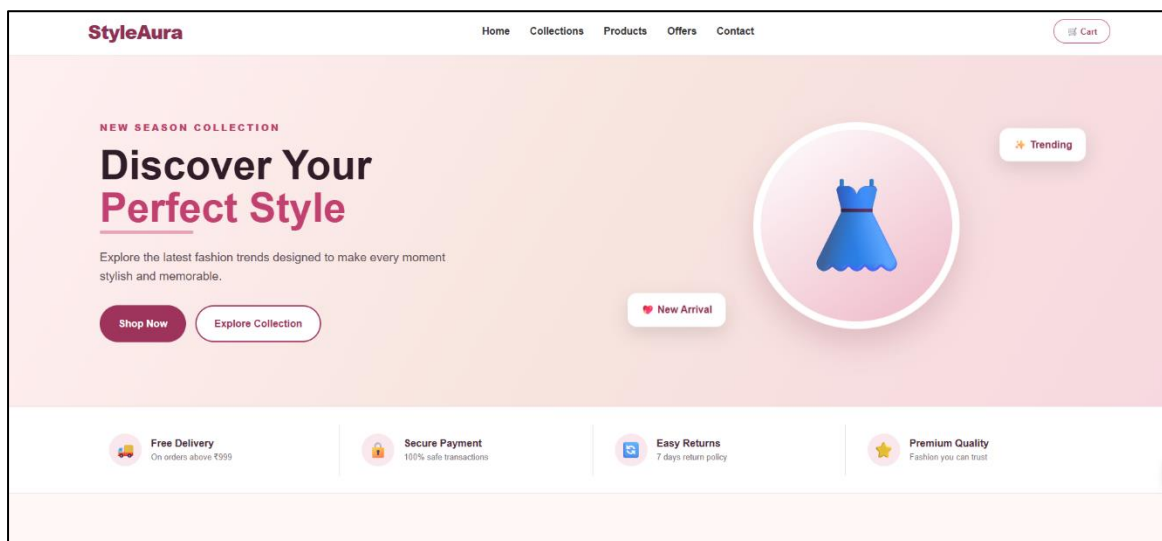
@keyframes appear {
  from {
    opacity: 0;
    transform: translateY(20px);
  }

  to {
    opacity: 1;
    transform: translateY(0);
  }
}

@media (max-width: 800px) {
  .products {
    grid-template-columns: repeat(2, 1fr);
  }
}

```

Output:



EXPLORE

Shop By Category

Find your perfect look from our latest collections.



Women's Fashion

Dresses, tops, jeans and more

Explore →



Men's Fashion

Shirts, jackets, trousers and more

Explore →



Accessories

Bags, watches, jewellery and more

Explore →



Footwear

Sneakers, heels and casual shoes

Explore →

OUR COLLECTION

Trending Products

Discover the styles everyone is talking about.

NEW



WOMEN

Elegant Floral Dress

★★★★★

₹1,499 ₹2,499

Add to Cart

SALE



MEN

Premium Casual Shirt

★★★★★

₹999 ₹1,699

Add to Cart

TRENDING



ACCESSORIES

Classic Designer Bag

★★★★★

₹1,799 ₹2,499

Add to Cart

-30%



FOOTWEAR

Urban Casual Sneakers

★★★★★

₹1,299 ₹1,899

Add to Cart

LIMITED TIME OFFER

Up to 50% OFF

Fashion For Every Occasion

Refresh your wardrobe with our exclusive seasonal collection.

Shop Sale Collection →



Stay in Style

Subscribe to receive new arrivals, fashion tips and exclusive offers.

Enter your email address

Subscribe

StyleAura

Fashion that reflects your personality.

Quick Links

Home

Collections

Discounts

Customer Service

+91 98765 43210

support@styleaura.com

Follow Us

Instagram Facebook Pinterest

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Explanation of Code

- The HTML structure contains a header, navigation bar, hero section, and product-category cards.
- The grid layout creates a multiple-column layout for the fashion categories.
- Background colors, gradients, borders, rounded corners, and spacing are used to improve the visual appearance.
- Text effects such as letter-spacing and text-transform are applied to headings.
- The @keyframes rule creates a CSS animation when the product cards appear.
- The :hover effect creates an interactive UI when the user moves the mouse over a product card.
- The @media rule makes the layout responsive on smaller screens.

Conclusion:

The online fashion-store homepage was successfully designed using HTML5 and CSS3. Different CSS properties were used to create attractive backgrounds, borders and text effects. CSS animations and hover effects improve user interaction and engagement. The multiple-column and responsive layout provides a user-friendly shopping interface

4.Scenario: A school wants to automate the student grade calculation process.

Task: Develop a JavaScript program that accepts marks of five subjects and displays the Total, Average, Grade, and Pass/Fail status. The solution should demonstrate:

- Variables and Data Types
- Literals
- Operators
- Statements (Selection and Iteration)
- User-defined Functions

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Grade Calculator</title>
</head>
```

```
<body>
```

```
<h1>Student Grade Calculator</h1>
```

```
<input type="number" id="m1" placeholder="Subject 1">
```

```
<input type="number" id="m2" placeholder="Subject 2">
```

```
<input type="number" id="m3" placeholder="Subject 3">
```

```
<input type="number" id="m4" placeholder="Subject 4">
```

```
<input type="number" id="m5" placeholder="Subject 5">
```

```
<button onclick="calculate()">Calculate Result</button>
```

```
<div id="result"></div>
```

```
<script>
```

```
// User-defined function to calculate the result
```

```
function calculate() {
```

```
    let a = Number(document.getElementById("m1").value);
```

```
    let b = Number(document.getElementById("m2").value);
```

```
    let c = Number(document.getElementById("m3").value);
```

```
    let d = Number(document.getElementById("m4").value);
```

```
    let e = Number(document.getElementById("m5").value);
```

```
    // Calculate total and average
```

```
    let total = a + b + c + d + e;
```

```
    let average = total / 5;
```

```
    // Determine grade
```

```
    let grade;
```

```
    if (average >= 90)
```

```
        grade = "A+";
```

```
    else if (average >= 80)
```

```
        grade = "A";
```

```
    else if (average >= 70)
```

```
        grade = "B";
```

```
    else if (average >= 60)
```

```
        grade = "C";
```

```
    else
```

```
        grade = "F";
```

```

// Determine pass or fail
let status =
  (a >= 40 && b >= 40 && c >= 40 &&
   d >= 40 && e >= 40)
  ? "PASS" : "FAIL";

// Display result
document.getElementById("result").innerHTML =
  "Total: " + total + "<br>" +
  "Average: " + average.toFixed(2) + "<br>" +
  "Grade: " + grade + "<br>" +
  "Status: " + status}

</script>

</body>
</html>

```

Output:

Student Grade Calculator

Enter marks for five subjects to calculate your result

Student Name

Enter Subject Marks

Subject 1

Subject 2

Subject 3

Subject 4

Subject 5

Calculate Result

Reset

Student Result

Name: Shobana

Total

477 / 500

Average

95.40

Grade

A+

Status

PASS

Explanation of Code

- let variables are used to store the marks of five subjects.
- Number() converts the entered values into numbers.
- Arithmetic operators are used to calculate the total and average.
- if...else if...else statements are used to determine the grade.
- Logical operators are used to check whether the student has passed all subjects.
- The user-defined calculate() function performs the complete grade calculation.
- innerHTML is used to display the Total, Average, Grade and Pass/Fail status.

Conclusion:

The student grade calculator was successfully implemented using JavaScript. It accepts marks for five subjects and calculates the total and average. Conditional statements are used to determine the grade and pass/fail status. The program demonstrates variables, operators, statements, and a user-defined function as required.

5. Scenario: A supermarket wants to generate customer bills automatically.

Task: Develop a JavaScript program to calculate the bill amount for three purchased products. The

program should:

- Accept quantity and unit price for each product
- Calculate the total bill
- Apply a 10% discount if the total amount exceeds ₹2000
- Display the final payable amount

The solution should demonstrate:

- Variables and Data Types
- Literals
- Operators

- Statements
- User-defined Functions

Program:

```
// Function to calculate the bill

function calculateBill(quantity, price) {

    // Calculate amount for each product

    let amount = quantity * price;

    return amount;

}

// Product details

let quantity1 = 2;

let price1 = 500;

let quantity2 = 3;

let price2 = 300;

let quantity3 = 2;

let price3 = 250;

// Calculate total bill

let total =

    calculateBill(quantity1, price1) +

    calculateBill(quantity2, price2) +

    calculateBill(quantity3, price3);

// Apply 10% discount if total exceeds ₹2000

let discount = 0;

if (total > 2000) {

    discount = total * 0.10;
```

```

}

// Calculate final payable amount

let finalAmount = total - discount;

console.log("Total Bill: ₹" + total);

console.log("Discount: ₹" + discount);

console.log("Final Payable Amount: ₹" + finalAmount);

```

Output:

Supermarket Billing System

Enter product details to generate your bill

Product Details

01	Product Name Laptop	Quantity 2	Unit Price (₹) 50001
02	Product Name Samsung Mobile	Quantity 1	Unit Price (₹) 40000
03	Product Name Remote	Quantity 1	Unit Price (₹) 500

Generate Bill

Reset

Bill Summary			
Product	Qty	Unit Price	Amount
Laptop	2	₹50001.00	₹100002.00
Samsung Mobile	1	₹40000.00	₹40000.00
Remote	1	₹500.00	₹500.00
Subtotal			₹140502.00
Discount			₹14050.20
Final Payable Amount			₹126451.80

Supermarket Billing System • JavaScript

Explanation of Code:

- Variables are used to store the quantity and unit price of three products.
- The calculateBill() function calculates the amount for each product.
- Arithmetic operators are used to calculate the total bill.
- An if statement checks whether the total exceeds **₹2000**.
- A **10% discount** is applied when the condition is satisfied.
- The final payable amount is calculated by subtracting the discount from the total.

Conclusion:

The supermarket billing system was successfully implemented using JavaScript. It calculates the bill for three products using their quantity and unit price. A 10% discount is automatically applied when the bill exceeds ₹2000. The final payable amount is then calculated and displayed.